MA388 Sabermetrics: Lesson 26

Statast Spray Charts

```
library(tidyverse)
library(knitr)
library(baseballr)
```

Review

Last class we talked about the streakiness or clumpiness coefficient. What does this metric measure and how do we calculate it?

True or False. If a player had a high batting average for the season, this means that they were not a streaky player.

Could a player be temporally streaky but spatially predictable?

What do Joey Gallo, Albert Pujols, and DJ LeMahieu have in common?

Here's one answer: They had some of the most extreme spray charts among MLB hitters in 2017.

Of course, all three hitters went to different extremes – Gallo was one of MLB's most pull-heavy hitters, Pujols hit balls to center field more than almost anyone, and LeMahieu used the opposite field like nobody else in baseball.

Here is the full article: Extreme pull and opposite-field hitters (MLB.com, 2017)

Let's explore their spray charts for that year.

```
library(tidyverse)
library(baseballr)
```

```
gallo <- scrape_statcast_savant(start_date = "2017-05-01",
                                 end_date = "2017-12-31",
                                 playerid = 608336,
                                 player_type = "batter")

pujols <- scrape_statcast_savant(start_date = "2017-05-01",
                                 end_date = "2017-12-31",
                                 playerid = 405395,
                                 player_type = "batter")

lemahieu <- scrape_statcast_savant(start_date = "2017-05-01",
                                 end_date = "2017-12-31",
                                 playerid = 518934,
                                 player_type = "batter")

all_data = rbind(gallo, lemahieu, pujols)
```

Filter the data to only include balls in play.

```
all_bip <- all_data |>
  filter(type == "X")

# hc_x and hc_y are the hit coordinates for balls in play.
all_bip |>
  select(player_name, events, hc_x, hc_y) |>
  head()
```

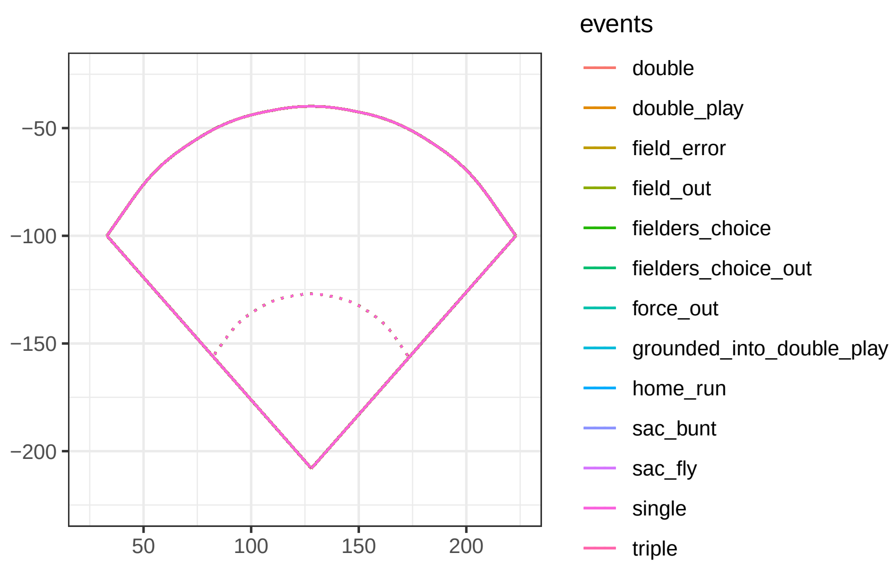
```
   player_name    events  hc_x   hc_y
        <char>    <char> <num>  <num>
1: Gallo, Joey  home_run 115.47  14.79
2: Gallo, Joey  home_run 227.17  67.05
3: Gallo, Joey field_out 156.70 154.25
4: Gallo, Joey    single 126.94  85.99
5: Gallo, Joey field_out 154.54  74.44
6: Gallo, Joey field_out 170.77 144.09
```

Now let's create the base plot for our spray chart.

```
spray_chart <- function(...) {
  ggplot(...) +
    geom_curve(x = 33, xend = 223, y = -100, yend = -100,
               curvature = -.65) +
    geom_segment(x = 128, xend = 33, y = -208, yend = -100) +
    geom_segment(x = 128, xend = 223, y = -208, yend = -100) +
    geom_curve(x = 83, xend = 173, y = -155, yend = -156,
               curvature = -.65, linetype = "dotted") +
    coord_fixed() +
    scale_x_continuous(NULL, limits = c(25, 225)) +
    scale_y_continuous(NULL, limits = c(-225, -25))
}
```

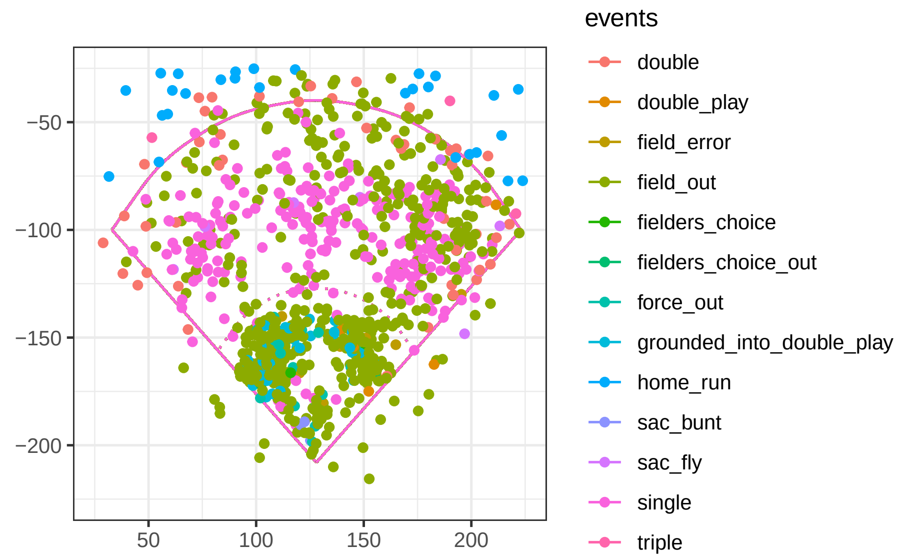
Where would we expect these different events to occur?

```
spray_chart(all_bip, aes(x = hc_x, y = -hc_y, color = events)) +
  scale_color_hue()+
  theme_bw()
```



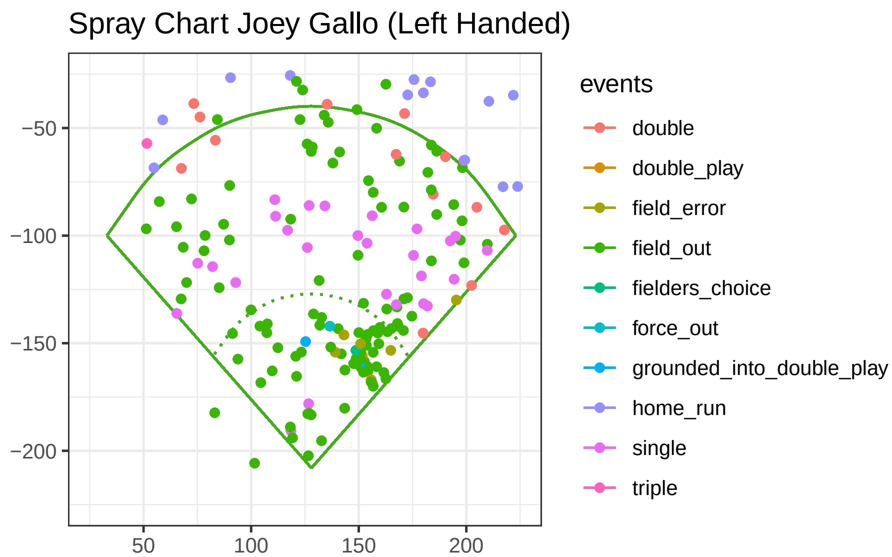
Let's look at all three players together.

```
spray_chart(all_bip, aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  theme_bw()
```

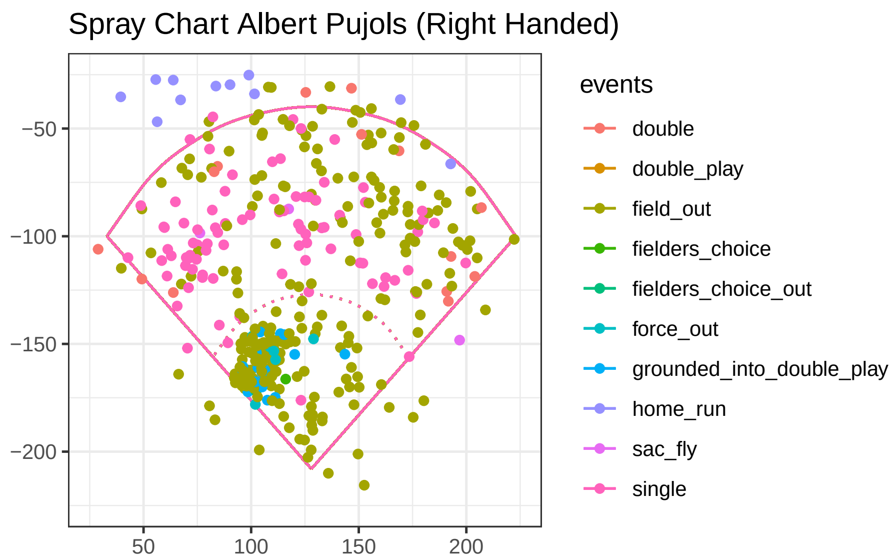


Now let's take a look at each player individually.

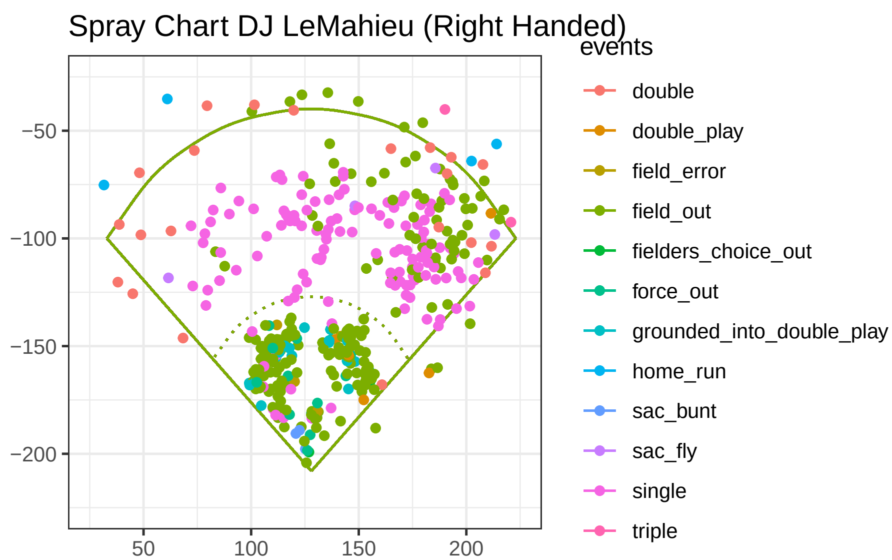
```
all_bip |>
  filter(player_name=="Gallo, Joey") |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  labs(title = "Spray Chart Joey Gallo (Left Handed)")+
  theme_bw()
```




```
all_bip |>
  filter(player_name=="Pujols, Albert") |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  labs(title = "Spray Chart Albert Pujols (Right Handed)")+
  theme_bw()
```

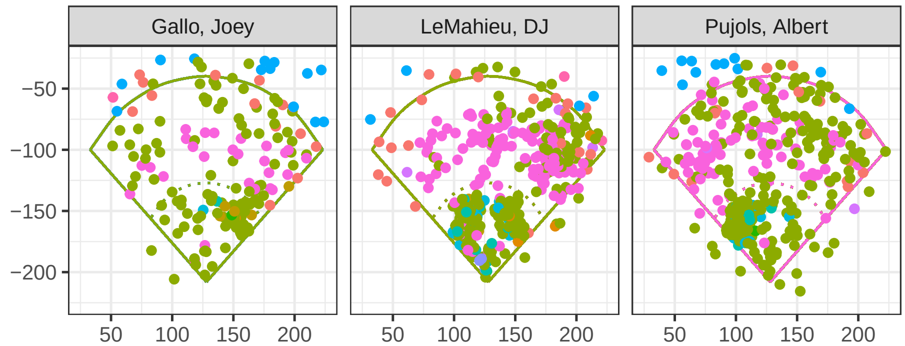


```
all_bip |>
  filter(player_name=="LeMahieu, DJ") |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  labs(title = "Spray Chart DJ LeMahieu (Right Handed)")+
  theme_bw()
```



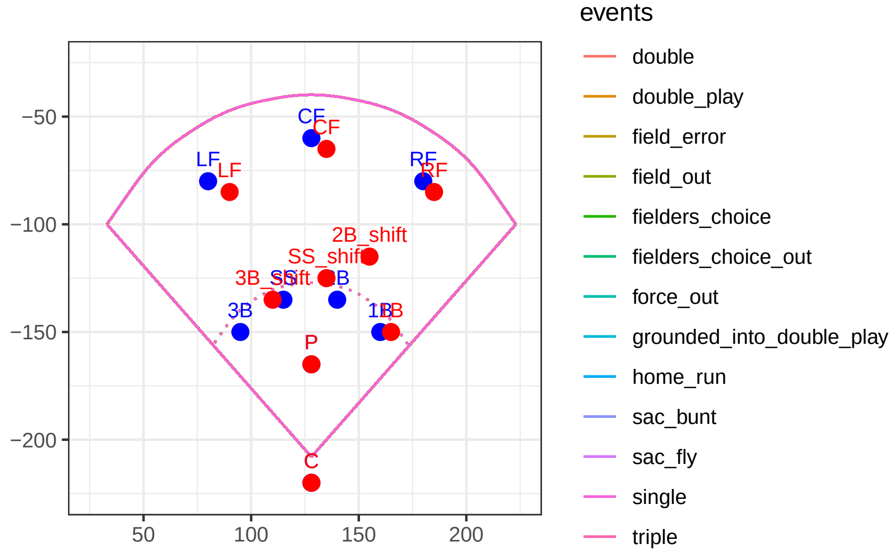
For a side-by-side comparison:

```
all_bip |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  theme_bw()+
  theme(legend.position = "none")+
  facet_wrap(~player_name)
```




```
# Traditional defensive alignment
traditional_defense <- tibble(
  position = c("P", "C", "1B", "2B", "SS", "3B", "LF", "CF", "RF"),
  x = c(128, 128, 160, 140, 115, 95, 80, 128, 180),
  y = c(-165, -220, -150, -135, -135, -150, -80, -60, -80)
)

# Shift vs pull-heavy LEFTY (e.g., Gallo)
shift_defense <- tibble(
  position = c("P", "C", "1B", "2B_shift", "SS_shift", "3B_shift", "LF", "CF", "RF"),
  x = c(128, 128, 165, 155, 135, 110, 90, 135, 185),
  y = c(-165, -220, -150, -115, -125, -135, -85, -65, -85)
)

add_defense <- function(defense_data, color = "black") {
  list(
    geom_point(data = defense_data,
               aes(x = x, y = y),
               color = color,
               size = 3),

    geom_text(data = defense_data,
              aes(x = x, y = y, label = position),
              vjust = -1,
              color = color,
              size = 3)
```

```r
  )
}

all_bip |>
  # filter(player_name == "Gallo, Joey") |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  # geom_point(alpha = 0.6) +
  add_defense(traditional_defense, color = "blue") +
  add_defense(shift_defense, color = "red") +
  # labs(title = "Joey Gallo vs Traditional Defense") +
  theme_bw()
```



In general, players hit ground balls to the pull side of the field. Let's take a look at how these three players hit their ground balls and the percentage of times they faced infield shifts.

```r
left_handed_batters = gallo |>
  filter(launch_angle < 10) |>
  summarize(N = n(),
            Pulled = 100 * mean(hc_x > 125.42, na.rm = TRUE),
            Shift = 100 *
              mean(if_fielding_alignment == "Infield shift",
                   na.rm = TRUE),.by='player_name')

right_handed_batters = all_data |>
  filter(player_name!="Gallo, Joey") |>
```

```r
  filter(launch_angle < 10) |>
  summarize(N = n(),
            Pulled = 100 * mean(hc_x < 125.42, na.rm = TRUE),
            Shift = 100 *
              mean(if_fielding_alignment == "Infield shift",
                   na.rm = TRUE),.by='player_name')

rbind(left_handed_batters, right_handed_batters)
```

```
# A tibble: 3 x 4
  player_name         N Pulled Shift
  <chr>           <int>  <dbl> <dbl>
1 Gallo, Joey        93   91.5  87.1
2 LeMahieu, DJ      312   49.8   0
3 Pujols, Albert    240   83.0  40
```

What does this tell us about the tendencies of these batters and how defenses approached them?